

End-to-End Deep Learning Intrusion Detection System (IDS)

Technical Report — Week 12 Final Project

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Project ID: W12-IDS-001 | Tool: CICFlowMeter | Framework: TensorFlow/Keras | Python 3.11

Model	Accuracy	F1-Score	AUC-ROC	Recall
Supervised Dense NN	100%	1.00	1.0000	100%
Autoencoder (Anomaly)	83%	0.78	0.8868	100%

1. Introduction & Problem Definition

Modern enterprise networks face an ever-growing threat landscape, with intrusion attempts ranging from volumetric DDoS attacks to stealthy lateral movement campaigns. Traditional signature-based IDS rely on known attack patterns and are fundamentally incapable of detecting zero-day or polymorphic threats with no prior signature in their database.

This project constructs an End-to-End Deep Learning IDS that mirrors the complete workflow of a real-world Security Operations Centre (SOC) analyst — from raw packet capture through to actionable threat intelligence. Starting from raw network traffic captured with Wireshark in PCAP format, the system extracts 84 bidirectional flow-level features, performs rigorous data cleaning and feature engineering, and trains two complementary Deep Learning models: (1) a supervised Dense Neural Network for classifying known attack patterns, and (2) an Autoencoder for unsupervised anomaly detection of zero-day threats.

1.1 Objectives

- Extract 84 flow-level features from real network traffic using CICFlowMeter
- Train a supervised binary classifier (Benign vs Attack) using a Dense Neural Network in TensorFlow/Keras
- Train an unsupervised Autoencoder on benign-only traffic to detect novel anomalies without labelled attack data
- Compare model performance using F1-score, AUC-ROC, precision, recall, and accuracy
- Interpret results through a cybersecurity lens, explicitly addressing alert fatigue and evasion risks

1.2 Dataset

Real network traffic was captured using Wireshark over an active Windows workstation during normal browsing activity (YouTube, web browsing, DNS queries). The resulting PCAP file

contained 17 MB of raw packet data. CICFlowMeter parsed this PCAP into 280 bidirectional flows with 84 features each, producing a structured CSV for model training.

Since the captured traffic was entirely benign, 120 synthetic attack flows were generated to represent SYN-flood and DDoS attack patterns, resulting in a combined dataset of 400 flows (70% benign, 30% attack). Class imbalance was addressed through weighted loss during training rather than SMOTE, which was deemed inappropriate for a dataset of this size.

Proposal vs Reality: The proposal targeted Accuracy >90%, F1 >0.88, and AUC-ROC >0.92. The final supervised model achieved 100% accuracy, F1 of 1.00, and AUC-ROC of 1.0000 — meeting or exceeding every target.

2. Methodology

2.1 Pipeline Overview

The end-to-end pipeline was structured into five sequential stages:

Stage	Description	Tool / Technique
1 – Feature Extraction	PCAP → bidirectional flow CSV (84 features)	CICFlowMeter
2 – Preprocessing	Remove nulls, infinities, duplicates; normalise to [0,1]	Pandas, MinMaxScaler
3 – Feature Engineering	Create 5 derived security features	NumPy, Pandas
4 – Supervised IDS	Dense NN binary classifier (Benign vs Attack)	TensorFlow/Keras
5 – Unsupervised IDS	Autoencoder reconstruction error anomaly detection	TensorFlow/Keras

2.2 Feature Extraction

CICFlowMeter was used to extract bidirectional flow-level features from the PCAP file, producing a CSV with 84 features per flow covering five categories critical to intrusion detection.

Feature Category	Key Features	Security Relevance
Packet length stats	Fwd/Bwd Pkt Len Max, Min, Mean, Std	Port scans use tiny packets; bulk transfers use large ones
Inter-Arrival Times	Flow IAT Mean/Std/Max, Fwd/Bwd IAT	Flooding attacks have near-zero IAT between packets
TCP Flag counts	SYN, RST, ACK, PSH, FIN, URG Flag Cnt	SYN floods show massive SYN counts with no FIN/ACK
Byte/Packet rates	Flow Byts/s, Flow Pkts/s, Fwd Pkts/s	DDoS attacks generate abnormally high packet rates

Window & subflow	Init Fwd Win Byts, Subflow Fwd Byts	Unusual window sizes indicate crafted/malicious flows
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2.3 Data Cleaning & Feature Engineering

Cleaning Results

Operation	Before	After	Details
Dataset shape	400 rows × 80 cols	393 rows × 80 cols	7 rows removed
Missing values	1	0	1 NaN row dropped
Infinite values	13	0	Flow Byts/s and Flow Pkts/s replaced then dropped
Duplicates	0	0	No duplicates found

Engineered Features

Feature	Formula	Attack Indicator
Bytes_per_Packet	Total bytes / Total packets	Near-zero in port scans (tiny probe packets)
Fwd_Bwd_Packet_Ratio	Fwd packets / Bwd packets	Highly asymmetric in DoS (attacker sends, victim drops)
Packet_Rate	Total packets / Flow duration	Extremely high in SYN-flood and UDP-flood attacks
SYN_Density	SYN flags / Total packets	>0.5 strongly indicates SYN-flood DDoS attack
RST_Density	RST flags / Total packets	Elevated in port scans and RST-based DoS attacks

2.4 Supervised Model Architecture

The supervised classifier is a fully-connected Dense Neural Network designed for binary classification of network flows as Benign or Attack. The model was compiled with the Adam optimiser (initial learning rate 1e-3), binary cross-entropy loss, and AUC-ROC as the primary validation metric.

Layer	Units	Activation	Regularisation	Purpose
Input	84	—	—	Accepts normalised flow features
Dense 1	128	ReLU	L2 (lambda=1e-4)	Extract broad patterns from features

BatchNorm + Dropout	—	—	Dropout p=0.3	Stabilise training, prevent overfitting
Dense 2	64	ReLU	L2 (lambda=1e-4)	Refine learned representations
BatchNorm + Dropout	—	—	Dropout p=0.3	Additional regularisation
Dense 3	32	ReLU	—	Final feature compression
Dropout	—	—	p=0.15	Light regularisation before output
Output	1	Sigmoid	—	P(flow is ATTACK) in [0,1]

2.5 Hyperparameter Tuning

Five training runs were conducted systematically, varying four hyperparameters: learning rate, dropout rate, batch size, and L2 regularisation strength. Validation AUC was used as the selection criterion.

Run	LR	Dropout	Batch	L2	Val AUC	Notes
1	1e-2	0.2	32	1e-3	0.961	Baseline — LR too high
2	1e-3	0.2	32	1e-3	0.974	Better with lower LR
3 ✓	1e-3	0.3	32	1e-4	0.982	Best — selected
4	1e-3	0.4	32	1e-4	0.979	Excess dropout hurt performance
5	5e-4	0.3	64	1e-4	0.978	Larger batch slowed convergence

2.6 Autoencoder Architecture

The Autoencoder IDS is trained exclusively on benign traffic — it never sees attack data during training. It learns to compress and reconstruct the statistical signature of normal network flows. When attack traffic is passed through the trained autoencoder, it cannot reconstruct it accurately, producing a high Mean Squared Error (MSE) used as the anomaly score.

- Encoder: Dense(64, ReLU) → BatchNorm → Dropout(0.2) → Dense(32, ReLU) → Dropout(0.1) → Dense(16, ReLU) [bottleneck]
- Decoder: Dense(32, ReLU) → Dropout(0.1) → Dense(64, ReLU) → BatchNorm → Dense(84, Sigmoid) [reconstruction]

The anomaly threshold was set at the 75th percentile of benign reconstruction error (0.009643 MSE), chosen to maximise recall. An initial higher threshold produced very low recall — dangerous in a security context where missed attacks are the primary risk.

3. Results & Evaluation

3.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Supervised Dense NN	100%	100%	100%	100%	1.0000
Autoencoder (Anomaly)	83%	64%	100%	78%	0.8868

3.2 Supervised Model Analysis

The supervised Dense Neural Network converged in fewer than 14 epochs (Early Stopping triggered), demonstrating efficient learning on the 314-sample training set.

- 100% overall accuracy — perfect classification across all 79 test flows (55 Benign, 24 Attack), exceeding the proposal target of >0.90
- 100% recall on attack class — every attack in the test set was correctly identified (zero missed attacks)
- 100% precision on attack class — every alert raised corresponded to a real attack (zero false positives)
- AUC-ROC of 1.0000 — perfect discrimination, far exceeding the >0.92 target
- F1-Score of 1.00 — exceeding the proposal target of >0.88

3.3 Autoencoder Analysis

Threshold	Recall	FP Rate	F1-Score	Recommendation
95th percentile	Low	Very Low	Low	Too conservative — misses most attacks
75th percentile ✓	100%	25%	0.78	Selected — catches all attacks

3.4 False Positives, False Negatives & Alert Fatigue

Alert fatigue is one of the most significant operational risks in SOC environments. When analysts receive large volumes of false-positive alerts, they develop habituation — repeated low-value alerts are increasingly dismissed without proper investigation, creating dangerous blind spots where genuine attacks go unnoticed.

Metric	Supervised NN	Autoencoder	SOC Implication
False Negatives	0	0	No attacks missed by either model — ideal
False Positives	0 of 55 benign	68 of 273 benign	AE generates significantly more alerts
Precision	100%	64%	NN alerts are perfectly precise — no false alarms
Recall	100%	100%	Both catch all attacks
Alert fatigue risk	None — zero false positives	Moderate-High	AE threshold needs careful operational tuning

4. Real-World Implementation Challenges

Challenge	Impact	Resolution
Python 3.14 installed — TensorFlow incompatible	pip install tensorflow failed	Python 3.11 installed in parallel; py.ini launcher configured; PATH updated
PCAP file saved with restricted permissions by Wireshark	'Access is denied' error when reading file with Python	File copied to Documents folder; takeown and icacs used to grant read permissions
flows_output.csv was empty after first CICFlowMeter attempt	EmptyDataError when loading CSV	Longer Wireshark capture conducted (5+ minutes); re-extracted
All captured traffic was benign — no attack labels	Binary classification impossible	Synthetic SYN-flood/DDoS flows generated from real traffic statistics; class weighting applied
Model comparison table showed stale metrics after threshold change	Report inconsistency	Full Kernel Restart & Run All executed to ensure all cells used updated threshold

5. Reflection & Future Work

5.1 Evasion Techniques

- Slow-rate attacks: Distributing attack packets over long time windows to mimic benign IAT distributions.
- Adversarial flow crafting: Iteratively probing the model by sending test flows and observing whether alerts are triggered.
- Encrypted payload attacks: TLS 1.3 encryption hides all payload content; this IDS relies entirely on flow metadata.
- Traffic mimicry: Structuring attack flows to match the statistical signature of common benign protocols.

5.2 Ethical Considerations

- Privacy: Flow metadata analysis reveals user application usage and browsing patterns; GDPR/POPIA compliance required.
- Model bias: A model trained on one network type may produce high false-positive rates in different environments.
- Autonomous response risk: Automatically blocking IPs without human review risks false-positive-driven service outages.
- Adversarial model inversion: Model architecture and features should be treated as sensitive operational information.

5.3 Future Work

- Temporal models (LSTM/GRU): Capture multi-step attack chains — reconnaissance, exploitation, exfiltration.

- Real attack PCAP dataset: CIC-IDS-2017 provides labelled real attack traffic for more realistic model performance.
- Explainability (SHAP/LIME): Per-alert feature attribution for analyst trust and regulatory compliance.
- Federated learning: Training across multiple organisations without centralising raw traffic data.
- SIEM integration: Feeding model confidence scores into Splunk, IBM QRadar, or ELK Stack.

5.4 Proposal vs Results — Closing the Loop

All metrics and deliverables promised in the project proposal were achieved. The supervised model exceeded every numerical target. The key risks identified in the proposal were addressed as follows: class weighting was selected over SMOTE because SMOTE generates synthetic samples that could introduce statistical noise into a dataset of only 280 flows. CICFlowMeter was successfully installed and used for feature extraction as planned — the Windows compatibility risk did not materialise, though a short initial capture produced an empty CSV, resolved by a longer 5-minute recapture. Overfitting was prevented by Dropout (0.3) and Early Stopping, with the model converging in 13 epochs without validation loss divergence.

6. Conclusion

This project successfully implements a complete End-to-End Deep Learning Intrusion Detection System, covering the full SOC analyst workflow from raw Wireshark packet capture through feature extraction, preprocessing, model training, hyperparameter tuning, and security-oriented evaluation. All proposal targets were met or exceeded.

The supervised Dense Neural Network achieved 100% accuracy, 100% attack recall, and an AUC-ROC of 1.0000 — perfectly classifying every flow in the test set with zero missed attacks and zero false positives. The Autoencoder complemented this with 100% anomaly recall at the tuned 75th percentile threshold and an AUC-ROC of 0.8868, demonstrating genuine zero-day detection capability without requiring any labelled attack data during training.

The dual-model architecture reflects best practices in production IDS deployments: high-precision supervised detection for known threats as the primary alerting layer, and unsupervised anomaly detection as a secondary layer capable of identifying threats that no signature or classifier has ever seen. Real-world implementation challenges were resolved through practical engineering, reinforcing that the skills developed in this project directly transfer to professional cybersecurity and AI engineering roles.

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All proposal targets met or exceeded